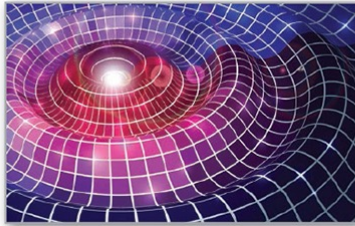


7 Eigenvalues And Eigenvectors



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1

7.1

Eigenvalues and Eigenvectors

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Objectives

- Verify eigenvalues and corresponding eigenvectors.
- Find eigenvalues and corresponding eigenspaces.
- Use the characteristic equation to find eigenvalues and eigenvectors, and find the eigenvalues and eigenvectors of a triangular matrix.
- Find the eigenvalues and eigenvectors of a linear transformation.

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The Eigenvalue problem

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The Eigenvalue problem (1 of 1)

Definitions of Eigenvalue and Eigenvector

Let A be an $n \times n$ matrix. The scalar λ is an **eigenvalue** of A when there is a *nonzero* vector $\mathbf{x}$ such that $A\mathbf{x} = \lambda\mathbf{x}$.

The vector $\mathbf{x}$ is an **eigenvector** of A corresponding to λ .

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Example 1 – Verifying Eigenvectors and Eigenvalues

For the matrix

$$A = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}$$

verify that $\mathbf{x}_1 = (1, 0)$ is an eigenvector of A corresponding to the eigenvalue $\lambda_1 = 2$, and that $\mathbf{x}_2 = (0, 1)$ is an eigenvector of A corresponding to the eigenvalue $\lambda_2 = -1$.

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Example 1 – Solution (1 of 2)

Multiplying $\mathbf{x}_1$ on the left by A produces

$$A\mathbf{x}_1 = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$



So, $\mathbf{x}_1 = (1, 0)$ is an eigenvector of A corresponding to the eigenvalue $\lambda_1 = 2$.

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Example 1 – Solution (2 of 2)

Similarly, multiplying $\mathbf{x}_2$ on the left by A produces

$$A\mathbf{x}_2 = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = -1 \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

So, $\mathbf{x}_2 = (0, 1)$ is an eigenvector of A corresponding to the eigenvalue $\lambda_2 = -1$.

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Eigenspaces

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Eigenspaces (1 of 1)

Theorem 7.1 Eigenvectors of λ Form a Subspace

If A is an $n \times n$ matrix with an eigenvalue λ , then the set of all eigenvectors of λ , together with the zero vector

$$\{\mathbf{x}: \mathbf{x} \text{ is an eigenvector of } \lambda\} \cup \{\mathbf{0}\}$$

is a subspace of R^n . This subspace is the **eigenspace** of λ .

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Example 3 – Finding Eigenspaces in $\mathbb{R}^2$ Geometrically

Find the eigenvalues and corresponding eigenspaces of

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Solution:

Geometrically, multiplying a vector (x, y) in $\mathbb{R}^2$ by the matrix A corresponds to a reflection in the y -axis. That is, if $\mathbf{v} = (x, y)$, then

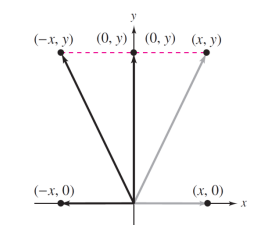
$$A\mathbf{v} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -x \\ y \end{bmatrix}.$$

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Example 3 – Solution (1 of 2)

Figure 7.1 illustrates that the only vectors reflected onto scalar multiples of themselves are those lying on either the x -axis or the y -axis.



A reflects vectors in the y -axis.

Figure 7.1

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Example 3 – Solution (2 of 2)

For a vector on the x -axis

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ 0 \end{bmatrix} = \begin{bmatrix} -x \\ 0 \end{bmatrix} = -1 \begin{bmatrix} x \\ 0 \end{bmatrix}$$

↑

Eigenvalue is $\lambda_1 = -1$.

For a vector on the y -axis

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ y \end{bmatrix} = 1 \begin{bmatrix} 0 \\ y \end{bmatrix}$$

↑

Eigenvalue is $\lambda_2 = 1$.

So, the eigenvectors corresponding to $\lambda_1 = -1$ are the nonzero vectors on the x -axis, and the eigenvectors corresponding to $\lambda_2 = 1$ are the nonzero vectors on the y -axis.

This implies that the eigenspace corresponding to $\lambda_1 = -1$ is the x -axis, and that the eigenspace corresponding to $\lambda_2 = 1$ is the y -axis.

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Finding Eigenvalues And Eigenvectors

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Finding Eigenvalues and Eigenvectors (1 of 3)

The equation $\det(\lambda I - A) = 0$ is the **characteristic equation** of A . Moreover, when expanded to polynomial form, the polynomial

$$|\lambda I - A| = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_2\lambda^2 + c_1\lambda + c_0$$

is the **characteristic polynomial** of A . So, the eigenvalues of an $n \times n$ matrix A correspond to the roots of the characteristic polynomial of A .

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Finding Eigenvalues and Eigenvectors (2 of 3)

Finding Eigenvalues and Eigenvectors

Let A be an $n \times n$ matrix.

1. Form the characteristic equation $|\lambda I - A| = 0$. It will be a polynomial equation of degree n in the variable λ .
2. Find the real roots of the characteristic equation. These are the eigenvalues of A .
3. For each eigenvalue λ_i , find the eigenvectors corresponding to λ_i by solving the homogeneous system $(\lambda_i I - A)\mathbf{x} = \mathbf{0}$. This can require row reducing an $n \times n$ matrix. The reduced row-echelon form must have at least one row of zeros.

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Example 5 – Finding Eigenvalues and Eigenvectors

Find the eigenvalues and corresponding eigenvectors of

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

What is the dimension of the eigenspace of each eigenvalue?

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Example 5 – Solution (1 of 2)

The characteristic polynomial of A is

$$\begin{aligned} |\lambda I - A| &= \begin{vmatrix} \lambda - 2 & -1 & 0 \\ 0 & \lambda - 2 & 0 \\ 0 & 0 & \lambda - 2 \end{vmatrix} \\ &= (\lambda - 2)^3. \end{aligned}$$

So, the characteristic equation is $(\lambda - 2)^3 = 0$, and the only eigenvalue is $\lambda = 2$. To find the eigenvectors of $\lambda = 2$, solve the homogeneous linear system represented by $(2I - A)\mathbf{x} = \mathbf{0}$.

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Example 5 – Solution (2 of 2)

$$2I - A = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

This implies that $x_2 = 0$. Using the parameters $s = x_1$ and $t = x_3$, you can conclude that the eigenvectors of $\lambda = 2$ are of the form

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} s \\ 0 \\ t \end{bmatrix} = s \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, s \text{ and } t \text{ not both zero.}$$

$\lambda = 2$ has two linearly independent eigenvectors, so the dimension of its eigenspace is 2.

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Finding Eigenvalues and Eigenvectors (3 of 3)

Theorem 7.3 Eigenvalues Of Triangular Matrices

If A is an $n \times n$ triangular matrix, then its eigenvalues are the entries on its main diagonal.

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Example 7 – Finding eigenvalues of Triangular and Diagonal matrices

Find the eigenvalues of each matrix.

a. $A = \begin{bmatrix} 2 & 0 & 0 \\ -1 & 1 & 0 \\ 5 & 3 & -3 \end{bmatrix}$

b. $A = \begin{bmatrix} -1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -4 & 0 \\ 0 & 0 & 0 & 0 & 3 \end{bmatrix}$

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Example 7 – Solution

a. Without using Theorem 7.3,

$$|\lambda I - A| = \begin{vmatrix} \lambda - 2 & 0 & 0 \\ 1 & \lambda - 1 & 0 \\ -5 & -3 & \lambda + 3 \end{vmatrix} = (\lambda - 2)(\lambda - 1)(\lambda + 3).$$

So, the eigenvalues are $\lambda_1 = 2$, $\lambda_2 = 1$, and $\lambda_3 = -3$, which are the main diagonal entries of A .

b. In this case, use Theorem 7.3 to conclude that the eigenvalues are the main diagonal entries $\lambda_1 = -1$, $\lambda_2 = 2$, $\lambda_3 = 0$, $\lambda_4 = -4$, and $\lambda_5 = 3$.

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Eigenvalues and Eigenvectors of Linear Transformations

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Eigenvalues and Eigenvectors of Linear Transformations (1 of 1)

Eigenvalues and eigenvectors can also be defined in terms of linear transformations. A number λ is an **eigenvalue** of a linear transformation $T: V \rightarrow V$ when there is a nonzero vector $\mathbf{x}$ such that $T(\mathbf{x}) = \lambda\mathbf{x}$.

The vector $\mathbf{x}$ is an **eigenvector** of T corresponding to λ , and the set of all eigenvectors of λ (with the zero vector) is the **eigenspace** of λ .

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Example 8 – Finding Eigenvalues and Eigenspaces

Find the eigenvalues and a basis for each corresponding eigenspace of

$$A = \begin{bmatrix} 1 & 3 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}.$$

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Example 8 – Solution

$$\begin{aligned} |\lambda I - A| &= \begin{vmatrix} \lambda - 1 & -3 & 0 \\ -3 & \lambda - 1 & 0 \\ 0 & 0 & \lambda + 2 \end{vmatrix} \\ &= [(\lambda - 1)^2 - 9](\lambda + 2) \\ &= (\lambda - 4)(\lambda + 2)^2 \end{aligned}$$

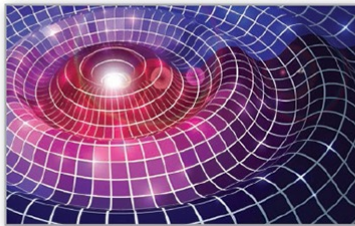
so the eigenvalues of A are $\lambda_1 = 4$ and $\lambda_2 = -2$. Bases for the eigenspaces are $B_1 = \{(1, 1, 0)\}$ and

$B_2 = \{(1, -1, 0), (0, 0, 1)\}$, respectively (verify these).

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7 Eigenvalues And Eigenvectors



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7.2 Diagonalization

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Objectives

- Find the eigenvalues of similar matrices, determine whether a matrix A is diagonalizable, and find a matrix P such that $P^{-1}AP$ is diagonal.
- Find, for a linear transformation $T: V \rightarrow V$, a basis B for V such that the matrix for T relative to B is diagonal.

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The Diagonalization Problem

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The Diagonalization Problem (1 of 6)

In this section, you will look at another classic problem in linear algebra called the **diagonalization problem**.

Definition of a Diagonalizable Matrix

An $n \times n$ matrix A is **diagonalizable** when A is similar to a diagonal matrix. That is, A is diagonalizable when there exists an invertible matrix P such that $P^{-1}AP$ is a diagonal matrix.

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The Diagonalization Problem (2 of 6)

Theorem 7.4 Similar Matrices Have the Same Eigenvalues

If A and B are similar $n \times n$ matrices, then they have the same eigenvalues.

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Example 2 – Finding Eigenvalues of Similar Matrices

The matrices A and D are similar.

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 1 \\ -1 & -2 & 4 \end{bmatrix} \quad \text{and} \quad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Use Theorem to find the eigenvalues of A .

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Example 2 – Solution

D is a diagonal matrix, so its eigenvalues are the entries on its main diagonal—that is, $\lambda_1 = 1$, $\lambda_2 = 2$, and $\lambda_3 = 3$.

Matrices A and D are similar, so you know from Theorem that A has the same eigenvalues.

Check this by showing that the characteristic polynomial of A is $|\lambda I - A| = (\lambda - 1)(\lambda - 2)(\lambda - 3)$.

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The Diagonalization Problem (3 of 6)

Theorem 7.5 Condition for Diagonalization

An $n \times n$ matrix A is diagonalizable if and only if it has n linearly independent eigenvectors.

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The Diagonalization Problem (4 of 6)

Steps for Diagonalizing a Square Matrix

Let A be an $n \times n$ matrix.

1. Find n linearly independent eigenvectors $\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_n$ for A (if possible) with corresponding eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$. If n linearly independent eigenvectors do not exist, then A is not diagonalizable.
2. Let P be the $n \times n$ matrix whose consist of these eigenvectors. That is, $P = [\mathbf{p}_1 \ \mathbf{p}_2 \ \dots \ \mathbf{p}_n]$.

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The Diagonalization Problem (5 of 6)

Steps for Diagonalizing a Square Matrix

3. The diagonal matrix $D = P^{-1}AP$ will have the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ on its main diagonal. Note that the order of the eigenvectors used to form P will determine the order in which the eigenvalues appear on the main diagonal of D .

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Example 4 – Diagonalizing a Matrix

Show that the matrix A is diagonalizable.

$$A = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 3 & 1 \\ -3 & 1 & -1 \end{bmatrix}$$

Then find a matrix P such that $P^{-1}AP$ is diagonal.

Solution:

The characteristic polynomial of A is

$$|\lambda I - A| = (\lambda - 2)(\lambda + 2)(\lambda - 3).$$

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Example 4 – Solution (1 of 2)

So, the eigenvalues of A are $\lambda_1 = 2$, $\lambda_2 = -2$, and $\lambda_3 = 3$. From these eigenvalues, you obtain the reduced row-echelon forms and corresponding eigenvectors below.

	→		Eigenvector
$2I - A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -1 & -1 \\ 3 & -1 & 3 \end{bmatrix}$	→	$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$
$-2I - A = \begin{bmatrix} -3 & 1 & 1 \\ -1 & -5 & -1 \\ 3 & -1 & -1 \end{bmatrix}$	→	$\begin{bmatrix} 1 & 0 & -\frac{1}{4} \\ 0 & 1 & \frac{1}{4} \\ 0 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}$
$3I - A = \begin{bmatrix} 2 & 1 & 1 \\ -1 & 0 & -1 \\ 3 & -1 & 4 \end{bmatrix}$	→	$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$

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Example 4 – Solution (2 of 2)

Form the matrix P whose columns are the eigenvectors just obtained.

$$P = \begin{bmatrix} -1 & 1 & -1 \\ 0 & -1 & 1 \\ 1 & 4 & 1 \end{bmatrix}$$

This matrix is nonsingular (check this), which implies that the eigenvectors are linearly independent and A is diagonalizable. So, it follows that

$$P^{-1}AP = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

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The Diagonalization Problem (6 of 6)

Theorem 7.6 Sufficient Condition for Diagonalization

If an $n \times n$ matrix A has n *distinct* eigenvalues, then the corresponding eigenvectors are linearly independent and A is diagonalizable.

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Example 7 – Determining Whether a matrix is Diagonalizable

Determine whether the matrix A is diagonalizable.

$$A = \begin{bmatrix} 1 & -2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & -3 \end{bmatrix}$$

Solution:

A is a triangular matrix, so its eigenvalues are the main diagonal entries $\lambda_1 = 1$, $\lambda_2 = 0$, and $\lambda_3 = -3$. Moreover, these three values are distinct, so you can conclude from Theorem that A is diagonalizable.

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Diagonalization And Linear Transformations

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Diagonalization And Linear Transformations (1 of 1)

In terms of linear transformations, the diagonalization problem can be stated as: For a linear transformation

$$T: V \rightarrow V,$$

does there exist a basis B for V such that the matrix for T relative to B is diagonal?

The answer is “yes” when the standard matrix for T is diagonalizable.

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Example 8 – Finding a basis

Let $T: R^3 \rightarrow R^3$ be the linear transformation represented by $T(x_1, x_2, x_3) = (x_1 - x_2 - x_3, x_1 + 3x_2 + x_3, -3x_1 + x_2 - x_3)$. If possible, find a basis B for R^3 such that the matrix for T relative to B is diagonal.

Solution:

The standard matrix for T is

$$A = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 3 & 1 \\ -3 & 1 & -1 \end{bmatrix}.$$

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Example 8 – Solution

From Example 4, you know that A is diagonalizable. So, the three linearly independent eigenvectors found in Example 4 can be used to form the basis B . That is,

$$B = \{(-1, 0, 1), (1, -1, 4), (-1, 1, 1)\}.$$

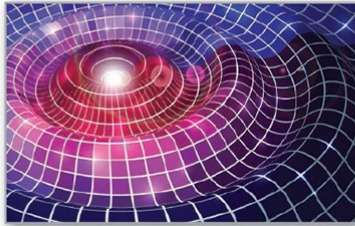
The matrix for T relative to this basis is

$$D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

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7 Eigenvalues And Eigenvectors



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7.3 Symmetric Matrices and Orthogonal Diagonalization

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Objectives

- Recognize, and apply properties of, symmetric matrices.
- Recognize, and apply properties of, orthogonal matrices.
- Find an orthogonal matrix P that orthogonally diagonalizes a symmetric matrix A .

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Symmetric Matrices

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Symmetric Matrices (1 of 2)

Definition of a Symmetric Matrix

A square matrix A is **symmetric** when it is equal to its transpose: $A = A^T$.

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Example 1 – Symmetric Matrices and non symmetric Matrices (1 of 2)

The matrices A and B are symmetric, but the matrix C is not.

$$A = \begin{bmatrix} 0 & 1 & -2 \\ 1 & 3 & 0 \\ -2 & 0 & 5 \end{bmatrix} \quad B = \begin{bmatrix} 4 & 3 \\ 3 & 1 \end{bmatrix} \quad C = \begin{bmatrix} 3 & 2 & 1 \\ 1 & -4 & 0 \\ 1 & 0 & 5 \end{bmatrix}$$

Non symmetric matrices have properties that are not exhibited by symmetric matrices, as listed below.

1. A non symmetric matrix may not be diagonalizable.
2. A non symmetric matrix can have eigenvalues that are not real.

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Example 1 – Symmetric Matrices and non symmetric Matrices (2 of 2)

3. For a non symmetric matrix, the number of linearly independent eigenvectors corresponding to an eigenvalue can be less than the multiplicity of the eigenvalue.

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Symmetric Matrices (2 of 2)**Theorem 7.7 Properties of Symmetric Matrices**

If A is an $n \times n$ symmetric matrix, then the properties listed below are true.

1. A is diagonalizable.
2. All eigenvalues of A are real.
3. If λ is an eigenvalue of A with multiplicity k , then λ has k linearly independent eigenvectors. That is, the eigenspace of λ has dimension k .

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Example 3 – Dimensions of the Eigenspaces of a Symmetric Matrix

Find the eigenvalues of the symmetric matrix

$$A = \begin{bmatrix} 1 & -2 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & -2 & 1 \end{bmatrix}$$

and determine the dimensions of the corresponding eigenspaces.

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Example 3 – Solution

The characteristic polynomial of A is

$$|\lambda I - A| = \begin{vmatrix} \lambda - 1 & 2 & 0 & 0 \\ 2 & \lambda - 1 & 0 & 0 \\ 0 & 0 & \lambda - 1 & 2 \\ 0 & 0 & 2 & \lambda - 1 \end{vmatrix} = (\lambda + 1)^2(\lambda - 3)^2.$$

So, the eigenvalues of A are $\lambda_1 = -1$ and $\lambda_2 = 3$. Each of these eigenvalues has a multiplicity of 2, so you know from Theorem that the corresponding eigenspaces also have dimension 2.

Specifically, the eigenspace of $\lambda_1 = -1$ has a basis of $B_1 = \{(1, 1, 0, 0), (0, 0, 1, 1)\}$ and the eigenspace of $\lambda_2 = 3$ has a basis of $B_2 = \{(1, -1, 0, 0), (0, 0, 1, -1)\}$.

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Orthogonal Matrices

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Orthogonal Matrices (1 of 2)

Definition of an Orthogonal Matrix

A square matrix P is **orthogonal** when it is invertible and $P^{-1} = P^T$.

THEOREM 7.8 Property of Orthogonal Matrices

An $n \times n$ matrix P is orthogonal if and only if its column vectors form an orthonormal set.

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Example 5 – An Orthogonal Matrix

Show that

$$P = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} & 0 \\ -\frac{2}{3\sqrt{5}} & -\frac{4}{3\sqrt{5}} & \frac{5}{3\sqrt{5}} \end{bmatrix}$$

is orthogonal by showing that $P^T = P^{-1}$. Then show that the column vectors of P form an orthonormal set.

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Example 5 – Solution (1 of 2)

$$PP^T = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} & 0 \\ -\frac{2}{3\sqrt{5}} & -\frac{4}{3\sqrt{5}} & \frac{5}{3\sqrt{5}} \end{bmatrix} \begin{bmatrix} \frac{1}{3} & -\frac{2}{\sqrt{5}} & -\frac{2}{3\sqrt{5}} \\ \frac{2}{3} & \frac{1}{\sqrt{5}} & -\frac{4}{3\sqrt{5}} \\ \frac{2}{3} & 0 & \frac{5}{3\sqrt{5}} \end{bmatrix} = I_3$$

so it follows that $P^T = P^{-1}$.

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Example 5 – Solution (2 of 2)

Moreover, letting

$$\mathbf{p}_1 = \begin{bmatrix} \frac{1}{3} \\ -\frac{2}{\sqrt{5}} \\ -\frac{2}{3\sqrt{5}} \end{bmatrix}, \mathbf{p}_2 = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{\sqrt{5}} \\ -\frac{4}{3\sqrt{5}} \end{bmatrix}, \text{ and } \mathbf{p}_3 = \begin{bmatrix} \frac{2}{3} \\ 0 \\ \frac{5}{3\sqrt{5}} \end{bmatrix}$$

produces $\mathbf{p}_1 \cdot \mathbf{p}_2 = \mathbf{p}_1 \cdot \mathbf{p}_3 = \mathbf{p}_2 \cdot \mathbf{p}_3 = 0$ and

$$\|\mathbf{p}_1\| = \|\mathbf{p}_2\| = \|\mathbf{p}_3\| = 1.$$

So, $\{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3\}$ is an orthonormal set, as guaranteed by Theorem.

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Orthogonal Matrices (2 of 2)

THEOREM 7.9 Property of Symmetric Matrices

Let A be an $n \times n$ symmetric matrix. If λ_1 and λ_2 are distinct eigenvalues of A , then their corresponding eigenvectors $\mathbf{x}_1$ and $\mathbf{x}_2$ are orthogonal.

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Example 6 – Eigenvectors of a Symmetric Matrix

Show that any two eigenvectors of

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

corresponding to distinct eigenvalues are orthogonal.

Solution:

The characteristic polynomial of A is

$$|\lambda I - A| = \begin{vmatrix} \lambda - 3 & -1 \\ -1 & \lambda - 3 \end{vmatrix} = (\lambda - 2)(\lambda - 4)$$

which implies that the eigenvalues of A are $\lambda_1 = 2$ and $\lambda_2 = 4$.

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Example 6 – Solution

Verify that every eigenvector corresponding to $\lambda_1 = 2$ is of the form

$$\mathbf{x}_1 = \begin{bmatrix} s \\ -s \end{bmatrix}, \quad s \neq 0$$

and every eigenvector corresponding to $\lambda_2 = 4$ is of the form

$$\mathbf{x}_2 = \begin{bmatrix} t \\ t \end{bmatrix}, \quad t \neq 0.$$

So, $\mathbf{x}_1 \cdot \mathbf{x}_2 = st - st = 0$

which means that $\mathbf{x}_1$ and $\mathbf{x}_2$ are orthogonal.

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Orthogonal Diagonalization

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Orthogonal Diagonalization (1 of 3)

A matrix A is **orthogonally diagonalizable** when there exists an orthogonal matrix P such that $P^{-1}AP = D$ is diagonal.

THEOREM 7.10 Fundamental Theorem of Symmetric Matrices

Let A be an $n \times n$ symmetric matrix. Then A is orthogonally diagonalizable (and has real eigenvalues) if and only if A is symmetric.

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Orthogonal Diagonalization (2 of 3)

Orthogonal Diagonalization of a Symmetric Matrix

Let A be an $n \times n$ symmetric matrix.

1. Find all eigenvalues of A and determine the multiplicity of each.
2. For *each* eigenvalue of multiplicity 1, find a unit eigenvector. (Find any eigenvector and then normalize it.)
3. For each eigenvalue of multiplicity $k \geq 2$, find a set of k linearly independent eigenvectors. (You know from Theorem 7.7 that this is possible.) If this set is not orthonormal, then apply the Gram-Schmidt orthonormalization process.

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Orthogonal Diagonalization (3 of 3)

Orthogonal Diagonalization of a Symmetric Matrix

4. The results of Steps 2 and 3 produce an orthonormal set of n eigenvectors.

Use these eigenvectors to form the columns of P . The matrix $P^{-1}AP = P^TAP = D$ will be diagonal. (The main diagonal entries of D are the eigenvalues of A .)

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Example 8 – Orthogonal Diagonalization

Find a matrix P that orthogonally diagonalizes

$$A = \begin{bmatrix} -2 & 2 \\ 2 & 1 \end{bmatrix}.$$

Solution

1. The characteristic polynomial of A is

$$|\lambda I - A| = \begin{vmatrix} \lambda + 2 & -2 \\ -2 & \lambda - 1 \end{vmatrix} = (\lambda + 3)(\lambda - 2).$$

So the eigenvalues are $\lambda_1 = -3$ and $\lambda_2 = 2$.

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Example 8 – Solution (1 of 3)

2. For each eigenvalue, find an eigenvector by converting the matrix $\lambda I - A$ to reduced row-echelon form.

$$\begin{aligned} -3I - A = \begin{bmatrix} -1 & -2 \\ -2 & -4 \end{bmatrix} &\rightarrow \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} -2 \\ 1 \end{bmatrix} \\ 2I - A = \begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix} &\rightarrow \begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 2 \end{bmatrix} \end{aligned}$$

Eigenvector

The eigenvectors $(-2, 1)$ and $(1, 2)$ form an *orthogonal* basis for $\mathbb{R}^2$.

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Example 8 – Solution (2 of 3)

Normalize these eigenvectors to produce an *orthonormal* basis.

$$\mathbf{p}_1 = \frac{(-2, 1)}{\|(-2, 1)\|} = \left(-\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right) \quad \mathbf{p}_2 = \frac{(1, 2)}{\|(1, 2)\|} = \left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$$

3. Each eigenvalue has a multiplicity of 1, so go directly to step 4.

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Example 8 – Solution (3 of 3)

4. Using $\mathbf{p}_1$ and $\mathbf{p}_2$ as column vectors, construct the matrix P .

$$P = \begin{bmatrix} -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{bmatrix}$$

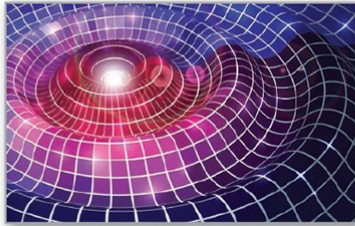
Verify that P orthogonally diagonalizes A by finding $P^{-1}AP = P^TAP$.

$$P^TAP = \begin{bmatrix} -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} -2 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{bmatrix} = \begin{bmatrix} -3 & 0 \\ 0 & 2 \end{bmatrix}$$

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7 Eigenvalues And Eigenvectors



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7.4

Applications of Eigenvalues and Eigenvectors

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Objectives

- Model population growth using an age transition matrix and an age distribution vector, and find a stable age distribution vector.
- Use a matrix equation to solve a system of first-order linear differential equations.
- Find the matrix of a quadratic form and use the Principal Axes Theorem to perform a rotation of axes for a conic and a quadric surface.
- Solve a constrained optimization problem.

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Population Growth

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Population Growth (1 of 1)

Matrices can be used to form models for population growth.

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Example 2 – Finding a Stable Age Distribution Vector

Find a stable age distribution vector for the population in rabbits has the characteristics below.

- a.** Half of the rabbits survive their first year. Of those, half survive their second year. The maximum life span is 3 years.
- b.** During the first year, the rabbits produce no offspring. The average number of offspring is 6 during the second year and 8 during the third year.

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Example 2 – Solution (1 of 2)

To solve this problem, find an eigenvalue and a corresponding eigenvector $\mathbf{x}$ such that $L\mathbf{x} = \lambda\mathbf{x}$. The characteristic polynomial of L is

$$|\lambda I - L| = (\lambda + 1)^2(\lambda - 2)$$

(check this), which implies that the eigenvalues are -1 and 2 . Choosing the positive value, let $\lambda = 2$. Verify that the corresponding eigenvectors are of the form

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 16t \\ 4t \\ t \end{bmatrix} = t \begin{bmatrix} 16 \\ 4 \\ 1 \end{bmatrix}.$$

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Example 2 – Solution (2 of 2)

For example, if $t = 2$, then the initial age distribution vector is

$$\mathbf{x}_1 = \begin{bmatrix} 32 \\ 8 \\ 2 \end{bmatrix} \quad \begin{array}{l} 0 \leq \text{age} < 1 \\ 1 \leq \text{age} < 2 \\ 2 \leq \text{age} \leq 3 \end{array}$$

and the age distribution vector for the next year is

$$\mathbf{x}_2 = L\mathbf{x}_1 = \begin{bmatrix} 0 & 6 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \end{bmatrix} \begin{bmatrix} 32 \\ 8 \\ 2 \end{bmatrix} = \begin{bmatrix} 64 \\ 16 \\ 4 \end{bmatrix}, \quad \begin{array}{l} 0 \leq \text{age} < 1 \\ 1 \leq \text{age} < 2 \\ 2 \leq \text{age} \leq 3 \end{array}$$

Notice that the ratio of the three age classes is still $16 : 4 : 1$, and so the percent of the population in each age class remains the same.

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Systems of Linear Differential Equations (Calculus)

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Systems of Linear Differential Equations (Calculus) (1 of 2)

A **system of first-order linear differential equations** has the form

$$y_1' = a_{11}y_1 + a_{12}y_2 + \cdots + a_{1n}y_n$$

$$y_2' = a_{21}y_1 + a_{22}y_2 + \cdots + a_{2n}y_n$$

$$\vdots$$

$$y_n' = a_{n1}y_1 + a_{n2}y_2 + \cdots + a_{nn}y_n$$

where each y_i is a function of t and $y_i' = \frac{dy_i}{dt}$.

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Systems of Linear Differential Equations (Calculus) (2 of 2)

If you let

$$\mathbf{y}' = \begin{bmatrix} y_1' \\ y_2' \\ \vdots \\ y_n' \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \text{and} \quad A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

then the system can be written in matrix form as

$$\mathbf{y}' = A\mathbf{y}.$$

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Example 4 – Solving a system of linear differential equations (1 of 3)

To solve the system of linear differential equations

$$\begin{aligned} y_1' &= 3y_1 + 2y_2 \\ y_2' &= 6y_1 - y_2 \end{aligned}$$

first find a matrix P that diagonalizes $A = \begin{bmatrix} 3 & 2 \\ 6 & -1 \end{bmatrix}$.

Verify that the eigenvalues of A are $\lambda_1 = -3$ and $\lambda_2 = 5$, and that the corresponding eigenvectors are

$$\mathbf{p}_1 = [1 \ -3]^T \text{ and } \mathbf{p}_2 = [1 \ 1]^T.$$

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Example 4 – Solving a system of linear differential equations (2 of 3)

Diagonalize A using the matrix P whose columns consist of $\mathbf{p}_1$ and $\mathbf{p}_2$ to obtain

$$P = \begin{bmatrix} 1 & 1 \\ -3 & 1 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} \frac{1}{4} & -\frac{1}{4} \\ \frac{3}{4} & \frac{1}{4} \end{bmatrix}, \quad \text{and} \quad P^{-1}AP = \begin{bmatrix} -3 & 0 \\ 0 & 5 \end{bmatrix}.$$

The system $\mathbf{w}' = P^{-1}AP\mathbf{w}$ has the form below.

$$\begin{bmatrix} w_1' \\ w_2' \end{bmatrix} = \begin{bmatrix} -3 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \quad \rightarrow \quad \begin{aligned} w_1' &= -3w_1 \\ w_2' &= 5w_2 \end{aligned}$$

The solution of this system of equations is

$$\begin{aligned} w_1 &= C_1 e^{-3t} \\ w_2 &= C_2 e^{5t}. \end{aligned}$$

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Example 4 – Solving a system of linear differential equations (3 of 3)

To return to the original variables y_1 and y_2 , use the substitution $\mathbf{y} = P\mathbf{w}$ and write

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

which implies that the solution is

$$\begin{aligned} y_1 &= w_1 + w_2 = C_1 e^{-3t} + C_2 e^{5t} \\ y_2 &= -3w_1 + w_2 = -3C_1 e^{-3t} + C_2 e^{5t}. \end{aligned}$$

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Quadratic Forms

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Quadratic Forms (1 of 11)

Eigenvalues and eigenvectors can be used to solve the rotation of axes problem. We know that classifying the graph of the quadratic equation

$$ax^2 + bxy + cy^2 + dx + ey + f = 0 \quad \text{Quadratic equation}$$

is fairly straightforward as long as the equation has no xy -term (that is, $b = 0$). If the equation has an xy -term, however, then the classification is accomplished most easily by first performing a rotation of axes that eliminates the xy -term.

The resulting equation (relative to the new $x'y'$ -axes) will then be of the form

$$a'(x')^2 + c'(y')^2 + d'x' + e'y' + f' = 0.$$

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Quadratic Forms (2 of 11)

You will see that the coefficients a' and c' are eigenvalues of the matrix

$$A = \begin{bmatrix} a & b/2 \\ b/2 & c \end{bmatrix}.$$

The expression

$$ax^2 + bxy + cy^2$$

Quadratic form

is the **quadratic form** associated with the quadratic equation

$$ax^2 + bxy + cy^2 + dx + ey + f = 0$$

and the matrix A is the **matrix of the quadratic form**. Note that the matrix A is *symmetric*. Moreover, the matrix A will be diagonal if and only if its corresponding quadratic form has no xy -term.

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Example 5 – Finding the Matrix of the Quadratic Form

Find the matrix of the quadratic form associated with each quadratic equation.

a. $4x^2 + 9y^2 - 36 = 0$ b. $13x^2 - 10xy + 13y^2 - 72 = 0$

Solution:

a. $a = 4$, $b = 0$, and $c = 9$, so the matrix is

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 9 \end{bmatrix}.$$

Diagonal matrix (no xy -term)

b. $a = 13$, $b = -10$, and $c = 13$, so the matrix is

$$A = \begin{bmatrix} 13 & -5 \\ -5 & 13 \end{bmatrix}.$$

Nondiagonal matrix (xy -term)

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Quadratic Forms (3 of 11)

Principal Axes Theorem

For a conic whose equation is $ax^2 + bxy + cy^2 + dx + ey + f = 0$, the rotation $X = PX'$ eliminates the xy -term when P is an orthogonal matrix, with $|P| = 1$, that diagonalizes the matrix of the quadratic form A . That is,

$$P^TAP = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

where λ_1 and λ_2 are eigenvalues of A . The equation of the rotated conic is

$$\lambda_1(x')^2 + \lambda_2(y')^2 + [d \ e]PX' + f = 0.$$

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Quadratic Forms (4 of 11)

The list below summarizes the steps used to apply the Principal Axes Theorem.

1. Form the matrix A and find its eigenvalues λ_1 and λ_2 .
2. Find eigenvectors corresponding to λ_1 and λ_2 . Normalize these eigenvectors to form the columns of P .
3. If $|P| = -1$, then multiply one of the columns of P by -1 to obtain a matrix of the form

$$P = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

4. The angle θ represents the angle of rotation of the conic.
5. The equation of the rotated conic is

$$\lambda_1(x')^2 + \lambda_2(y')^2 + [d \ e]PX' + f = 0.$$

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Example 7 – Rotation of a Conic

Perform a rotation of axes to eliminate the xy -term in the quadratic equation

$$3x^2 - 10xy + 3y^2 + 16\sqrt{2}x - 32 = 0.$$

Solution:

The matrix of the quadratic form associated with this equation is

$$A = \begin{bmatrix} 3 & -5 \\ -5 & 3 \end{bmatrix}.$$

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Example 7 – Solution (1 of 3)

The eigenvalues of A are $\lambda_1 = 8$ and $\lambda_2 = -2$ with corresponding eigenvectors of

$$\mathbf{x}_1 = (-1, 1) \text{ and } \mathbf{x}_2 = (-1, -1).$$

This implies that the matrix P is

$$P = \begin{bmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}, \text{ where } |P| = 1.$$

$\cos 135^\circ = -1/\sqrt{2}$ and $\sin 135^\circ = 1/\sqrt{2}$, so the angle of rotation is 135° .

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Example 7 – Solution (2 of 3)

Finally, from the matrix product

$$\begin{aligned} [d \ e]PX' &= [16\sqrt{2} \ 0] \begin{bmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} \\ &= -16x' - 16y' \end{aligned}$$

the equation of the rotated conic is

$$8(x')^2 - 2(y')^2 - 16x' - 16y' - 32 = 0.$$

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Example 7 – Solution (3 of 3)

In standard form, the equation is

$$\frac{(x' - 1)^2}{1^2} - \frac{(y' + 4)^2}{2^2} = 1$$

which is the equation of a hyperbola. Its graph is shown in Figure 7.4.

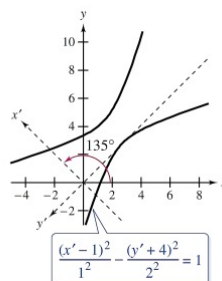


Figure 7.4

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Quadratic Forms (5 of 11)

There are six basic types of quadric surfaces: ellipsoids, hyperboloids of one sheet, hyperboloids of two sheets, elliptic cones, elliptic paraboloids, and hyperbolic paraboloids.

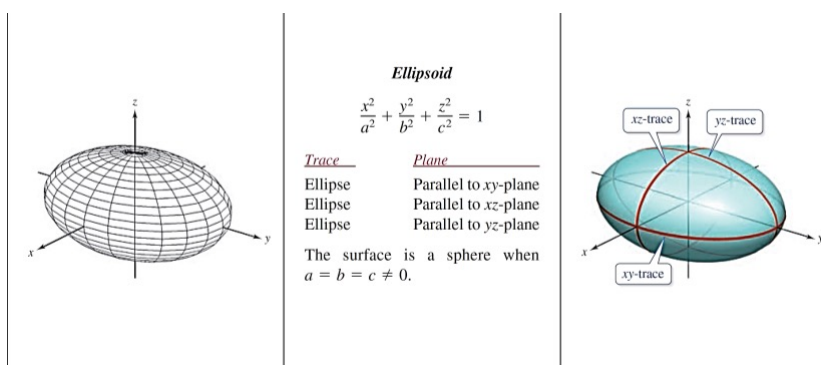
The intersection of a surface with a plane, called the **trace** of the surface in the plane, is useful to help visualize the graph of the surface in R^3 .

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Quadratic Forms (6 of 11)

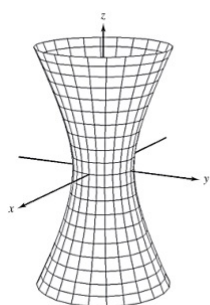
The six basic types of quadric surfaces, together with their traces, are shown.



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Quadratic Forms (7 of 11)



Hyperboloid of One Sheet

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

Trace

Ellipse

Hyperbola

Hyperbola

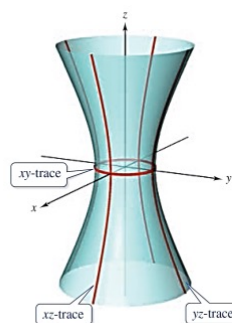
Plane

Parallel to xy -plane

Parallel to xz -plane

Parallel to yz -plane

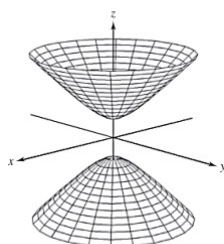
The axis of the hyperboloid corresponds to the variable whose coefficient is negative.



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Quadratic Forms (8 of 11)



Hyperboloid of Two Sheets

$$\frac{z^2}{c^2} - \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Trace

Ellipse

Hyperbola

Hyperbola

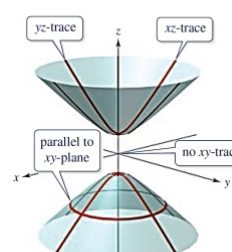
Plane

Parallel to xy -plane

Parallel to xz -plane

Parallel to yz -plane

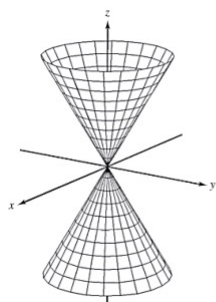
The axis of the hyperboloid corresponds to the variable whose coefficient is positive. There is no trace in the coordinate plane perpendicular to this axis.



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Quadratic Forms (9 of 11)



Elliptic Cone

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$$

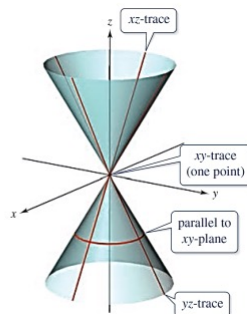
Trace

Ellipse
Hyperbola
Hyperbola

Plane

Parallel to xy -plane
Parallel to xz -plane
Parallel to yz -plane

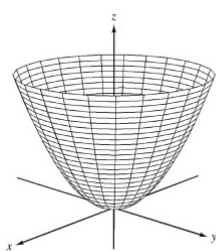
The axis of the cone corresponds to the variable whose coefficient is negative. The traces in the coordinate planes parallel to this axis are intersecting lines.



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Quadratic Forms (10 of 11)



Elliptic Paraboloid

$$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$$

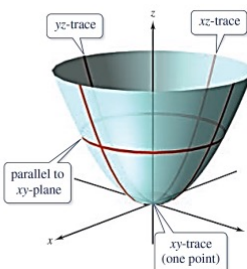
Trace

Ellipse
Parabola
Parabola

Plane

Parallel to xy -plane
Parallel to xz -plane
Parallel to yz -plane

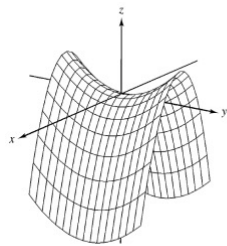
The axis of the paraboloid corresponds to the variable raised to the first power.



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Quadratic Forms (11 of 11)



Hyperbolic Paraboloid

$$z = \frac{y^2}{b^2} - \frac{x^2}{a^2}$$

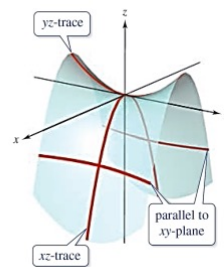
Trace

Hyperbola
Parabola
Parabola

Plane

Parallel to xy -plane
Parallel to xz -plane
Parallel to yz -plane

The axis of the paraboloid corresponds to the variable raised to the first power.



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Constrained Optimization

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Constrained Optimization (1 of 1)

Constrained optimization theorem

For a quadratic form f in n variables with matrix of the quadratic form A subject to the constraint $\|\mathbf{x}\|^2 = 1$, the maximum value of f is the greatest eigenvalue of A and the minimum value of f is the least eigenvalue of A .

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Example 10 – Finding Maximum and Minimum Values

Find the maximum and minimum values, and the corresponding normalized eigenvectors, of $z = 7x_1^2 + 6x_1x_2 + 7x_2^2$ subject to the constraint $\|\mathbf{x}\|^2 = 1$.

Solution:

The quadratic form f can be written using matrix notation as

$$f(x_1, x_2) = \mathbf{x}^T A \mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 7 & 3 \\ 3 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

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Example 10 – Solution

Verify that the eigenvalues of $A = \begin{bmatrix} 7 & 3 \\ 3 & 7 \end{bmatrix}$ are $\lambda_1 = 10$ and $\lambda_2 = 4$, with corresponding eigenvectors

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

So, the constrained maximum of 10 occurs when

$$(x_1, x_2) = \frac{1}{\sqrt{2}}(1, 1) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$$

and the constrained minimum of 4 occurs when

$$(x_1, x_2) = \frac{1}{\sqrt{2}}(-1, 1) = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right).$$

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